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Subject Name : Machine Learning

Unit No: 1

Unit Name : Introduction to Machine Learning & Data Preprocessing

Faculty Name : Dr. Ritu Jain

Syllabus of Machine Learning
Unit 1

Unit	Description	No of hours allotted	Marks % Weightage	CO

1	**Introduction to Machine Learning and Data Preprocessing**
	• Machine Learning: definition and need of ML,
	• Traditional Programming approach verses Machine Learning
	• Real-world applications of ML
	• Types of Machine Learning Systems:
	  o Supervised, Unsupervised, Reinforcement learning
	  o Batch and Online Learning
	  o Instance-Based and Model-Based Learning
	• Machine Learning Life Cycle
	• Data preprocessing in ML:
	  o Handling missing values
	  o Encoding categorical data
	  o Feature Scaling: Normalization and standardization
	  o Dataset splitting strategies:
	  ▪ Training, validation, and test sets,
	  ▪ K-Fold cross-validation
		7	16%	CO1

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Syllabus of Machine Learning
Unit 2

2	**Supervised Learning: Regression and Classification**
	• Linear regression (univariate & multivariate),
	• Cost functions and Gradient Descent (intuition)
	• Model evaluation concepts:
	  ○ Training error and generalization error
	  ○ Overfitting and underfitting diagnosis
	  ○ Bias–variance trade-off
	• Regularization concepts (L1, L2 – intuition)
	• Logistic Regression for classification (sigmoid, threshold)
	• Performance Metrics:
	  ○ Regression: MAE, MSE, RMSE, R^2
	  ○ Classification: Accuracy, Confusion Matrix, Precision, Recall, F1 Score, ROC Curve
		9	**20%**	**CO2**

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Syllabus of Machine Learning
Unit 3

3	**Tree-Based Models and Ensemble Learning**
	• Decision Trees
	  ○ Splitting Criteria:
	  ▪ Entropy (Information Gain)
	  ▪ Gini Index (conceptual)
	  ○ CART Algorithm (conceptual)
	  ○ Prunning Decision Trees
	• Ensemble Learning Techniques
	  ○ Bagging
	  ▪ Random Forest
	  ○ Boosting
	  ▪ AdaBoost (conceptual)
	  ▪ XGBoost (conceptual)
		8	**18%**	**CO3**

	• Hyperparameter Tuning: Grid Search (basic idea)			

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Syllabus of Machine Learning
Unit 4

4	**Support Vector Machines** • SVM Fundamentals: Margins, Support Vectors, Soft margin (C parameter) • Optimization and Kernel Trick (geometric understanding, RBF) • Linear SVM • Nonlinear SVM • Multiclass Classification (One-vs-One, One-vs-All)	**8**	**18%**	**CO4**

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Syllabus of Machine Learning
Unit 5

5	**Unsupervised Learning: Clustering**
	• Introduction to unsupervised learning
	• Distance and similarity measures ((Euclidean Distance, Manhattan Distance, Cosine Similarity)
	• Clustering techniques:
	  o K-Means (intuition, Elbow method, limitations)
	  o Hierarchical Clustering (agglomerative, dendrogram)
	  o DBSCAN (density-based intuition, ε, MinPts)
		5	**10%**	**CO5**

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Syllabus of Machine Learning
Unit 6

6	**Dimensionality Reduction and Model Deployment**
	• Curse of dimensionality
	• Dimensionality reduction techniques:
	  ○ Principal Component Analysis (PCA) (geometric understanding)
	  ○ Introduction to Linear Discriminant Analysis (LDA) (conceptual)
	• Feature selection vs feature extraction
	• Model persistence: Pickle vs Joblib
	• ML deployment: Flask APIs, Streamlit apps
	• Basic ML system design considerations
		8	18%	CO6

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Reference Books:

- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd Ed.) by Aurélien Géron
- Python Machine Learning (4th Ed.) by Sebastian Raschka
- Introduction to Machine Learning with Python by Andreas Mueller, Sarah Guido
- Pattern Recognition and Machine Learning by Christopher Bishop
- Machine Learning, McGraw-Hill, Tom M. Mitchell

Course Objectives

The objectives of this course are to:

1. Provide a strong foundation in the fundamental concepts of Machine Learning and essential data preprocessing techniques required for building predictive models.
2. Develop conceptual and practical understanding of supervised learning methods, including regression and classification, along with performance evaluation.
3. Introduce tree-based learning and ensemble techniques to enhance predictive performance and improve model reliability.
4. Familiarize with Support Vector Machine principles for handling linear and non-linear data.
5. Understand unsupervised learning approaches, particularly clustering and similarity-based analysis for discovering hidden patterns in data.
6. Introduce dimensionality reduction methods and basic model deployment practices, enabling students to understand modern machine learning workflows.

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Course Outcomes

After learning the course, the students should be able to:

1. Explain machine learning concepts and apply appropriate data preprocessing techniques.
2. Implement and evaluate supervised learning models for regression and classification problems using suitable performance metrics.
3. Analyze and apply decision tree and ensemble learning techniques to enhance model accuracy, generalization capability, and robustness.
4. Illustrate and apply the principles of Support Vector Machines for solving linear and non-linear classification problems.
5. Apply and interpret clustering techniques and similarity measures to discover patterns and structure in unlabelled datasets
6. Understand dimensionality reduction techniques and demonstrate basic skills in machine learning deployment.

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Unit No.: 1

Faculty Name :

Lecture No: 1

ARTIFICIAL INTELLIGENCE

- AI is a science that empowers computers to mimic human intelligence such as decision making, text processing, and visual perception, reasoning, and planning.
- It is a intelligence where we want to add all the capabilities to machine that human contain.
- AI is a broader field (i.e.: the big umbrella) that contains several subfield such as machine learning, robotics, and computer vision.

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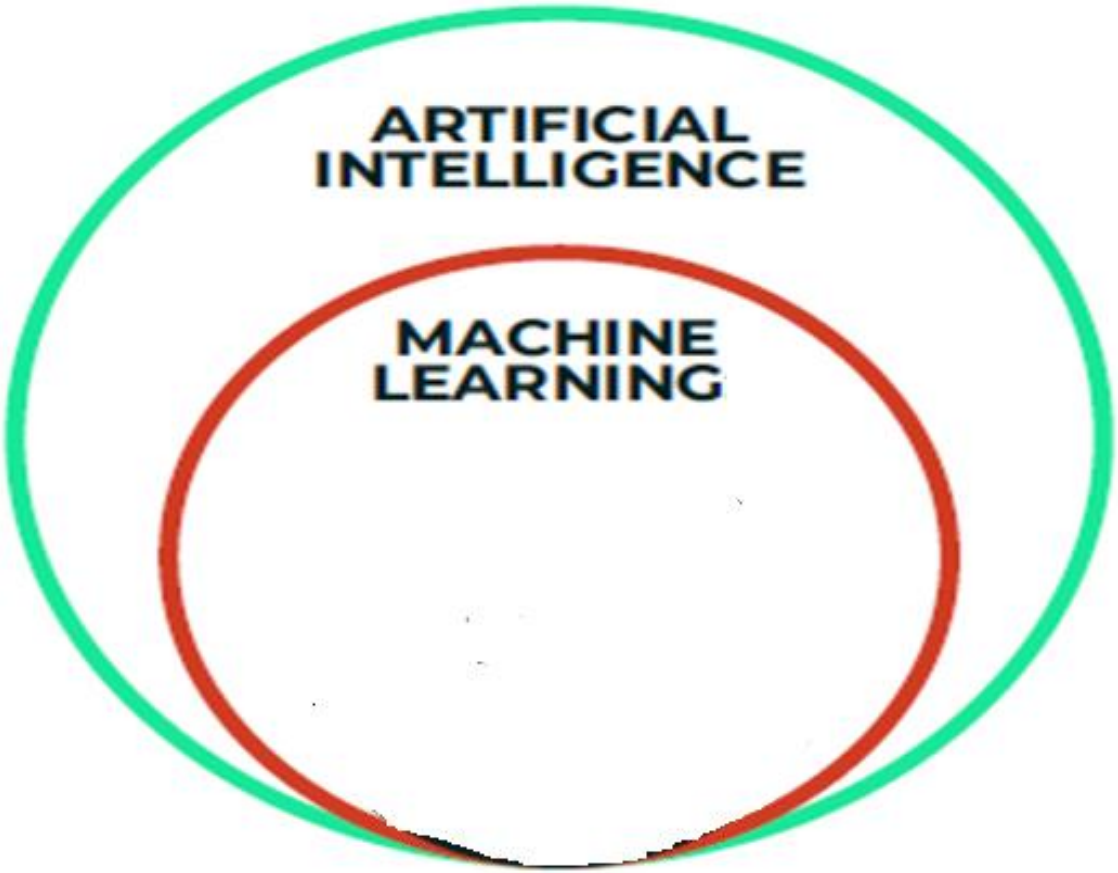
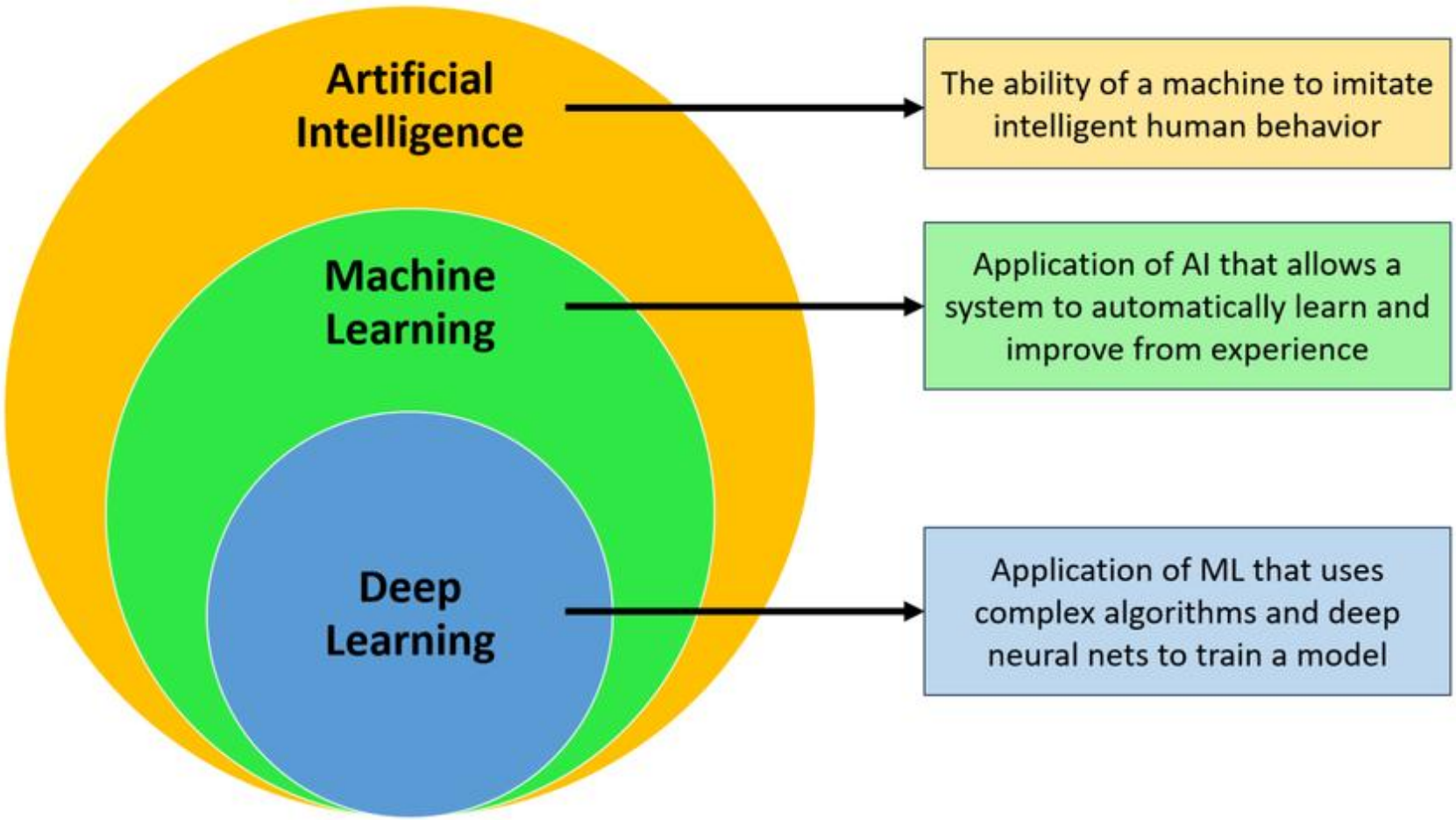
Relationship between AI, ML

- ML is seen as a subset of artificial intelligence.




Artificial Intelligence
Machine Learning
Deep Learning
The ability of a machine to imitate intelligent human behavior
Application of AI that allows a system to automatically learn and improve from experience
Application of ML that uses complex algorithms and deep neural nets to train a model

Definition of Machine Learning

- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.

  —Arthur Samuel, *1959*

- A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.
- —Tom Mitchell, *1997*

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Examples of MACHINE LEARNING

- **Handwriting recognition learning problem**

  Task T: Recognising and classifying handwritten words within images

  Performance P: Percent of words correctly classified

  Training experience E: A dataset of handwritten words with given classifications

- **Spam filter** is a Machine Learning program that can learn to flag spam given examples of spam emails (e.g., flagged by users) and examples of regular (nonspam, also called "ham") emails.
  - the **task T** is to flag spam for new emails,
  - the **experience E** is the *training data*, and
  - the per**formance measure P:** can be the ratio of correctly classified emails.

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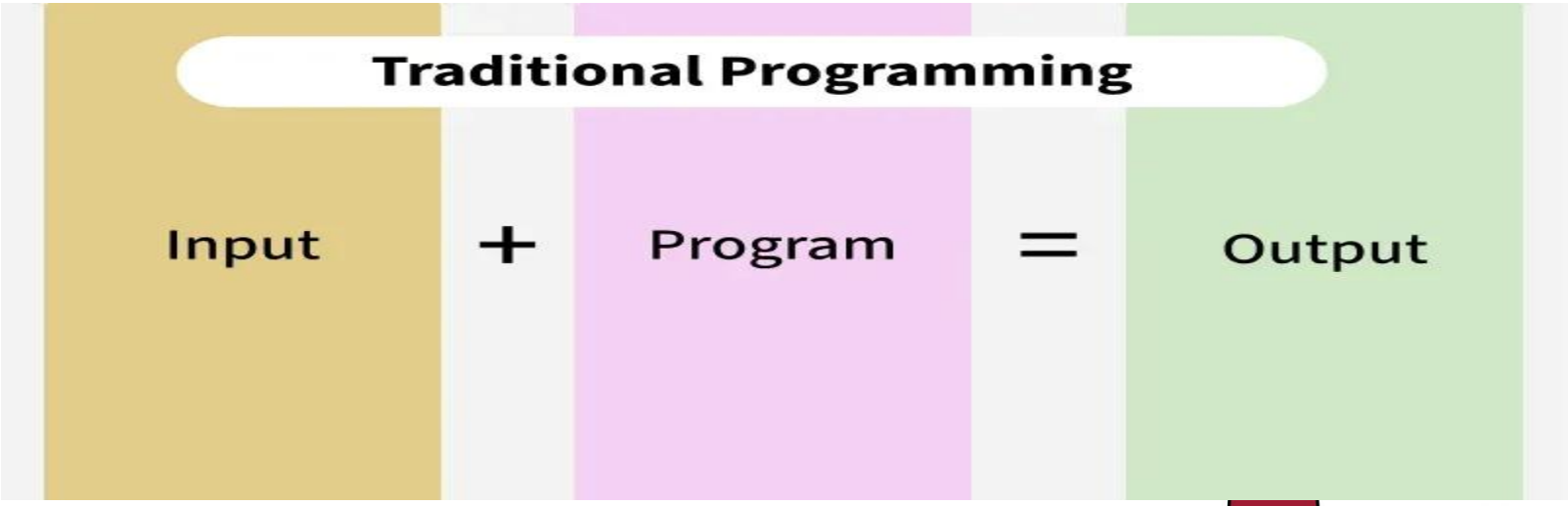
Traditional Programming

- **Traditional Programming:** It is a rule-based approach where developers write explicit instructions for the computer to follow.
- Every possible scenario must be anticipated and coded, making the logic transparent and deterministic.



Traditional Programming Approach

How Traditional Programming Works:

- **Input:** Data (usually structured and predictable)
- **Process:** Human-written rules and logic
- **Output:** Deterministic result (same input always yields the same output)

Characteristics of Traditional Programming Approach:

- **Rule-Based Logic:** The programmer defines every condition and outcome.
- **Deterministic Output:** The same input always produces the same result.
- **Manual Updates:** Any change in requirements or data format requires manual code changes.
- **Transparency:** The logic is clear, traceable, and easy to debug.
- **Limited Flexibility:** Struggles with complex or unpredictable scenarios.
- **Best For:** Well-defined, stable problems (e.g., payroll, calculators, sorting, transaction processing).

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Traditional Programming Approach

- ***Example***:
- *A spam filter built with traditional programming might use a list of banned words.*
- *The programmer writes rules: "If an email contains 'win money', mark as spam."*
- *Every new spam pattern requires a new rule.*

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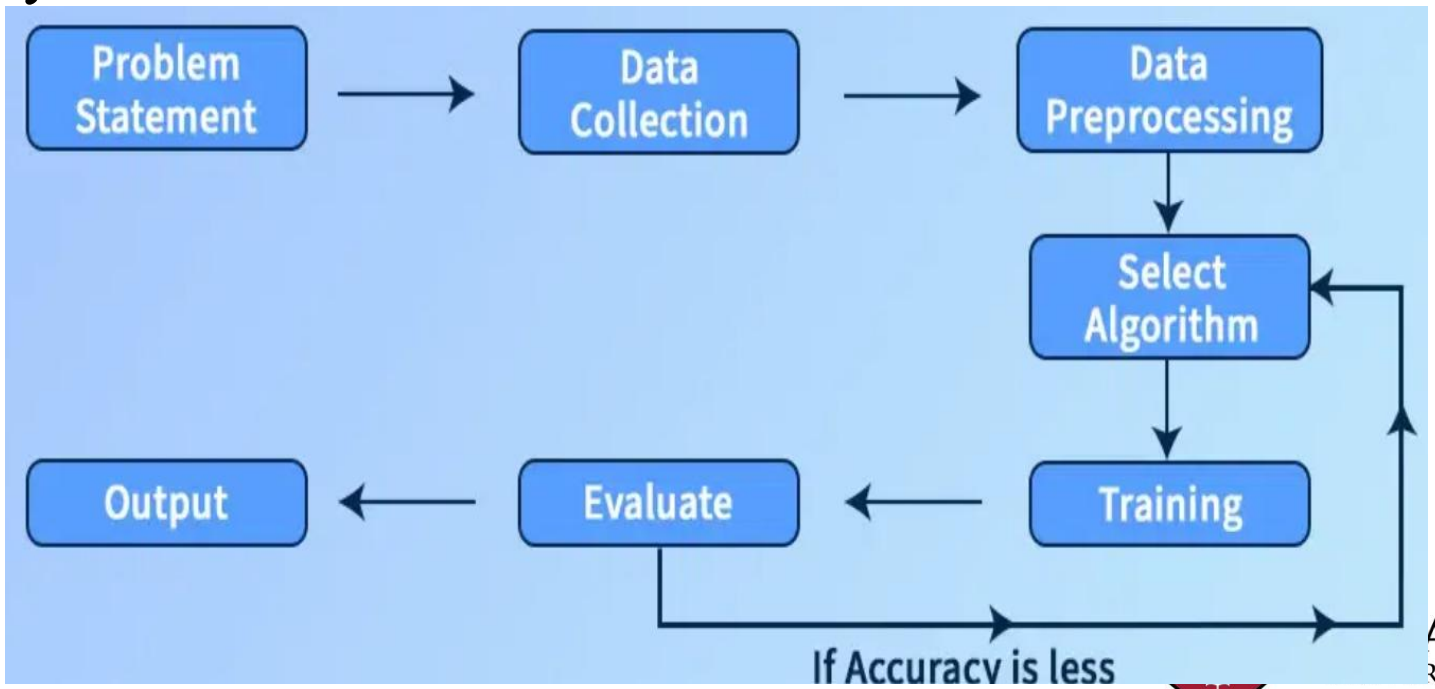
ML Approach

- **Machine Learning (ML):** It is a data-driven approach where algorithms learn patterns from data, enabling them to make predictions or decisions without explicit programming for every scenario.



ML Approach

How ML Works:

- **Input:** Large, often unstructured datasets (images, text, sensor data)
- **Process:** The algorithm learns from data during training, building a model that captures relationships
- **Output:** Probabilistic predictions or classifications (output may vary with new data)

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Characteristics of ML Approach:

- **Data-Driven Learning:** The model learns patterns from examples, not from explicit rules.
- **Probabilistic Output:** Results are predictions with associated probabilities, not fixed answers.
- **Adaptability:** The system can improve and adapt as new data arrives, often without manual intervention.
- **Opacity:** Complex models (like deep neural networks) can be difficult to interpret ("black box").
- **Best For:** Complex, data-rich problems where explicit rules are hard to define (e.g., image recognition, language translation, fraud detection).

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Example of ML Approach

- ***Example:***
- *A spam filter built with machine learning is trained on thousands of emails labeled as "spam" or "not spam."*
- *The model learns patterns in the data (word combinations, sender behavior, etc.) and can identify new spam emails.*

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Traditional Programming vs Machine Learning

- **Traditional programming**
  - relies on explicitly defined rules and logic written by developers to solve problems, making it ideal for tasks with clear, predictable outcomes.
- **Machine learning**
  - enables computers to learn patterns from data and make decisions or predictions, allowing them to handle complex or evolving tasks where manual rule writing is impractical.
- This key difference shapes how each approach is applied in real-world scenarios.

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Key Differences between Traditional Programming and ML

Factor	Traditional Programming	Machine Learning
Instruction Method	Explicit rules and logic	Learns patterns from data
Data Handling	Structured, predictable data	Large, often unstructured data
Outcome Predictability	Always the same result for same input	Predictions may vary as model adapts to new data
Flexibility	Limited to predefined conditions	Adjusts based on new data (self-improving)
Development Process	Linear: write, debug, deploy	Iterative: train, evaluate, tune, retrain
Transparency	Easy to trace and debug	Can be opaque, often needs explainable AI
Problem Complexity	Best for simple, well-defined tasks	Best for complex, data-rich tasks

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Why the Difference Matters

1. Flexibility & Adaptability

- **Traditional programming** is rigid. Changes in requirements or data require manual code updates.
- **Machine learning** adapts over time, making it ideal for dynamic environments like fraud detection or evolving business needs.

2. Problem Complexity

- **Rule-based systems** excel at straightforward tasks but struggle with complex problems like image recognition or language processing.
- **Machine learning** can handle these complex, multi-layered challenges by finding patterns in vast, diverse datasets.

3. Decision-Making & Predictability

- **Traditional programming** provides clear, consistent outputs.
- **Machine learning** offers probability based predictions, valuable in scenarios where patterns evolve or are hard to define.

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Why the Difference Matters

4. Transparency & Explainability

- **Traditional programming** is easy to explain every decision follows a visible rule.
- **Machine learning** can be opaque, especially with deep learning, so explainable AI is important for trust in sensitive applications (like finance or healthcare).

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Practical Implications

- **Traditional programming** is best for fixed, stable environments with clear rules and structured data. Example: Tax calculation, transaction processing, inventory management.
- **Machine learning** is essential for modern, data-rich, and dynamic and complex applications where explicit rules are impractical or impossible to define. Example : Voice assistants, recommendation engines, medical image analysis, autonomous vehicles.
  - It can also be used for **Fluctuating environments as it adapt to new data.** Example: Environmental & Climate Monitoring, Stock Market Trading
  - ML automates time-consuming, repetitive tasks with high accuracy hence reducing manual work and errors.
    - Gmail filtering spam emails automatically

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Unit No:1

Lecture No: 2

Need of Machine Learning Systems

- Machine learning (ML) systems are essential for processing massive datasets to uncover hidden patterns, automate complex, repetitive tasks, and enable real-time, data-driven decision-making.
- They are critical for enhancing operational efficiency, providing personalized user experiences, and advancing innovation in areas like autonomous vehicles, fraud detection, and predictive analytics. Voice assistants like Siri and Alexa learning our preferences and accents

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Key Reasons for the Need of Machine Learning Systems

- **Handling Big Data & Complexity:** Traditional programming cannot manage the sheer volume and complexity of data generated today. ML algorithms identify intricate trends within vast datasets, offering insights for business and research.
- **Automation and Efficiency**: ML automates low-cognition, repetitive tasks (e.g., data entry, classification, document processing), allowing for increased productivity.
- **Predictive Analytics**: Businesses utilize ML to predict future outcomes, such as customer behavior, market trends, or equipment failures, reducing risks and improving proactive decision-making.
- **Personalization**: ML powers recommendation engines, such as those used by Netflix, Amazon, and Spotify, by analyzing user history and browsing behavior to tailor content and

Key Reasons for the Need of Machine Learning Systems

- **Improved Security:** ML systems excel at detecting fraudulent behavior, identifying cybersecurity threats, and managing risk in real-time.
- **Advanced Capabilities:** Technologies like computer vision and natural language processing (NLP) enable sophisticated applications, including autonomous vehicles, voice assistants, and language translation.

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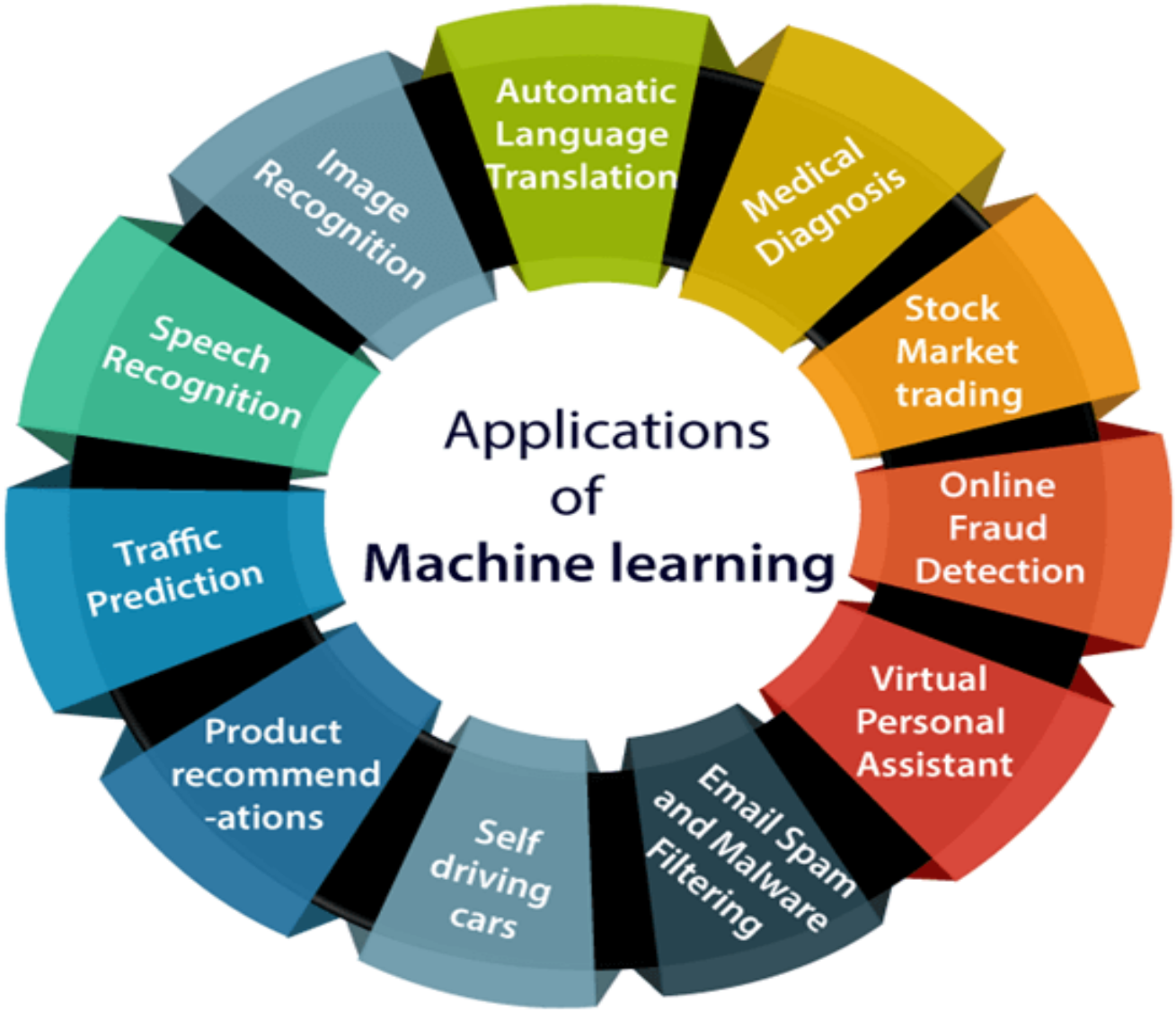
Applications of Machine Learning



Applications of Machine Learning

- **Healthcare and Medical Diagnosis:** ML algorithms can analyze large volumes of patient data, medical scans and genetic information to aid in diagnosis and treatment.
  - **Applications:**
    - **Disease Detection**: ML models are used to identify diseases like cancer, pneumonia and Parkinson's from medical images. They often achieve accuracy comparable to or better than human doctors.
    - **Predictive Analytics**: By analyzing patient history and symptoms, models can predict the risk of certain diseases or potential complications.
    - **Drug Discovery**: ML accelerates the drug development process by predicting how different compounds will interact, reducing the time and cost of research.

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Applications of Machine Learning

- **Smart Assistants and Human-Machine Interaction**
- Virtual assistants systems rely on **natural language processing_(NLP)** and **speech recognition** to understand commands and respond intelligently.
  - **Applications:**
    - **Voice Assistants**: Tools like Siri, Alexa and Google Assistant convert spoken input into actionable commands.
    - **Voice Search & Transcription**: ML enables users to perform hands-free web searches and get transcription during meetings or phone calls.
    - **Chatbots**: Businesses use AI-powered chatbots for 24/7 customer support, helping resolve queries faster and

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Applications of Machine Learning

- **Personalized Recommendations and User Experience**
- Modern digital platforms uses personalization which is done by using **recommender_systems**. Machine learning models analyze user behavior to deliver relevant content, improving engagement and satisfaction.
  - **Applications:**
    - **Streaming Platforms**: Netflix and Spotify suggest shows and songs based on your watching or listening history.
    - **E-commerce**: Sites like Amazon recommend products tailored to your preferences, browsing patterns and past purchases.
    - **Social Media**: Algorithms curate content feeds,

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Applications of Machine Learning

- **Fraud Detection and Financial Forecasting**
- In finance, vast sums of money move digitally and machine learning plays a important role in fraud detection and market analysis.
  - **Applications:**
    - **Transaction Monitoring**: Banks use ML models to detect unusual spending behavior and flag suspicious transactions.
    - **Loan Risk Assessment**: Credit scoring models analyze customer profiles and predict the likelihood of default.
    - **Stock Market Prediction**: ML is used to analyze historical stock data and forecast price movements. Stock markets are complex, algorithmic trading uses

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Applications of Machine Learning

- **Autonomous Vehicles and Smart Mobility**
  - Self-driving vehicles use ML to understand their environment, navigate safely and make immediate decisions.
    - **Key Components:**
    - **Computer Vision**: Recognizing lanes, pedestrians, traffic signals and obstacles.
    - **Sensor Fusion**: Combining data from cameras, LiDAR and radar for a 360-degree view.
    - **Behavior Prediction**: Anticipating how other drivers or pedestrians may act.
  - ML is also being used in traffic optimization, smart navigation systems and predictive maintenance in transportation.
  - **Traffic prediction:** Google Maps shows us the correct path with the shortest route and predicts the traffic conditions.

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Applications of Machine Learning

- **Image Recognition:**
- Image recognition is used to identify objects, persons, places, digital images, etc.
- The popular use case of image recognition and face detection is, Automatic friend tagging suggestion by Facebook :
  - Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's face detection and recognition algorithm.
  - It is based on the Facebook project named "Deep Face," which is responsible for face recognition and person identification in the picture.

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Applications of Machine Learning

- ***Email Spam and Malware Filtering:***
  Whenever we receive a new email, it is filtered automatically as important, normal, and spam.
- **Automatic Language Translation:** Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

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Types of Machine Learning Systems

- There are so many different types of Machine Learning systems that it is useful to classify them in broad categories based on:
  - *Whether or not they are trained with human supervision* **(supervised, unsupervised, and Reinforcement Learning)**
  - *Whether or not they can learn incrementally on the fly* **(online versus batch learning)**
  - *Whether they work by simply comparing new data points to known data points, or instead detect patterns in the training data and build a predictive model, much like scientists do* **(instance-based versus model-based learning)**
- These criteria are not exclusive; you can combine them in any way you like. For example, a spam filter may learn on the fly using a model trained using supervised learning; this makes it an online, model-based, supervised learning system.

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Unit No:1

Lecture No: 3

Type of ML Systems on the basis of amount of Supervision

- Machine Learning systems can be classified according to the amount and type of supervision they get during training.
- There are three major categories:
    1. Supervised Machine Learning
    2. Unsupervised Machine Learning
    3. Reinforcement Learning
- Additionally, there is a more specific category called **Semi-Supervised Learning** and **Self-Supervised Learning**, which combines elements of both supervised and unsupervised learning

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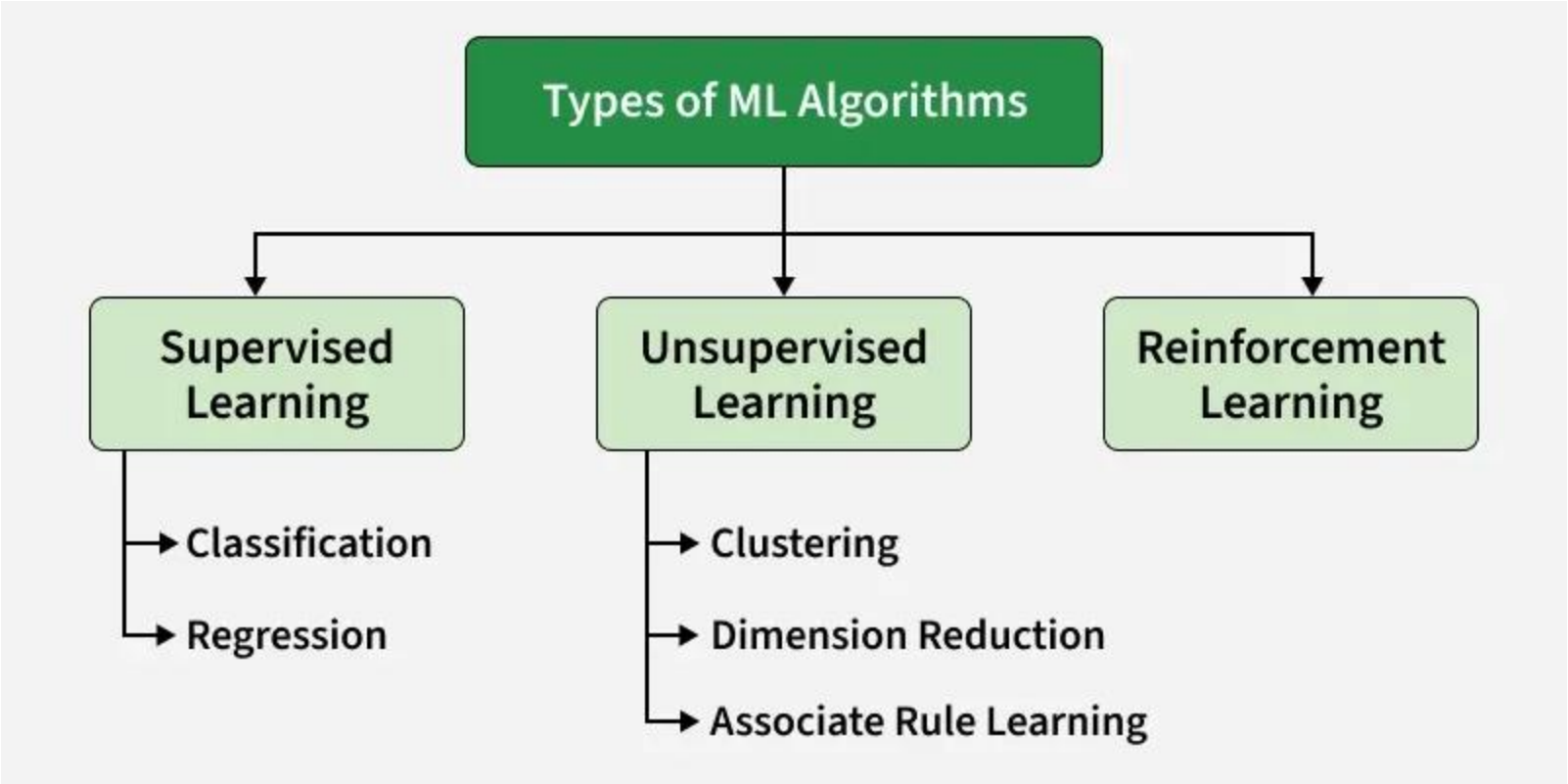
Type of ML Systems on the basis of amount of Supervision



Supervised Learning

- **Supervised learning** relies on **labeled datasets, where each input is paired with a corresponding output label**.
- Here, the goal of supervised machine learning algorithm is to learn the relationship between independent input features and a labeled target variable (dependent variable).
- In this type of machine learning algorithm,
  - The training data set is a labelled data set.
  - In other words, the training data set contains the input value (X) and target value (Y).
  - The learning algorithm generates a model.
  - Then, test data set consisting of only the input value is fed.
  - The model then predicts the target value based on its learning.

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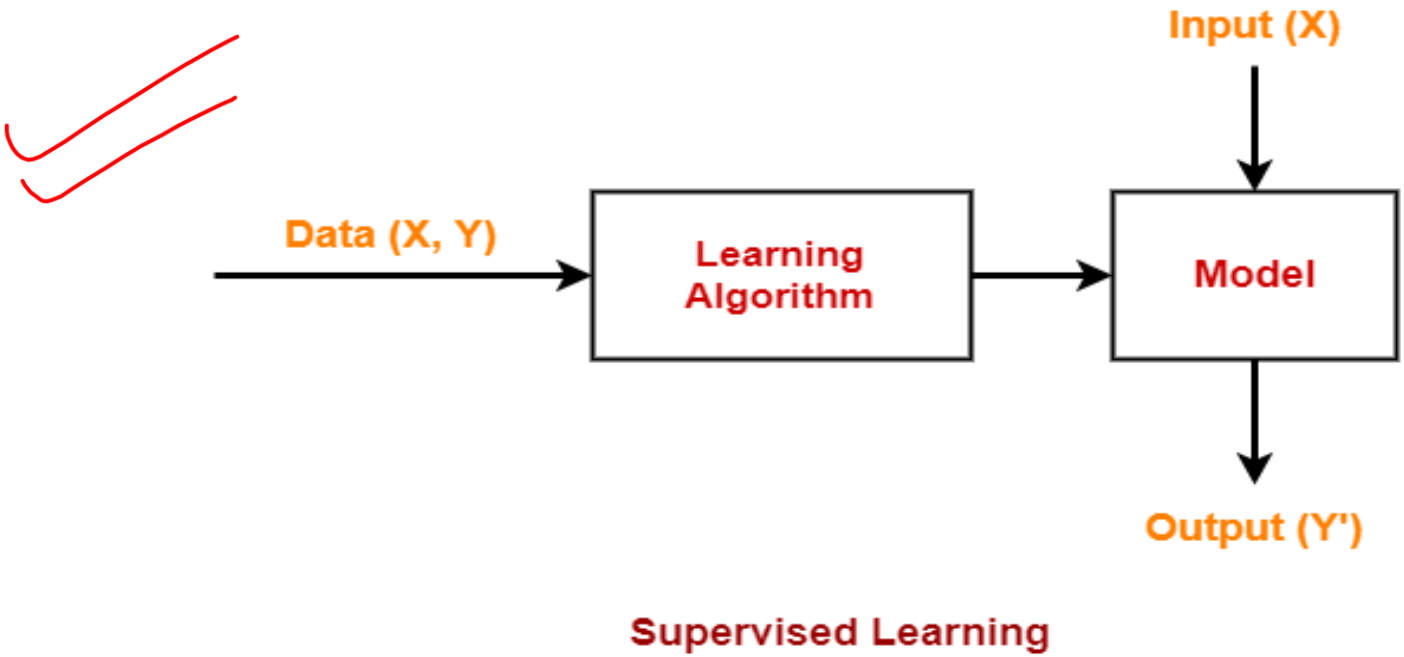
Supervised Learning



Supervised Learning Analogy:

•A teacher shows a child pictures of animals and labels them as "cat" or "dog.“ The child learns to recognize the features that distinguish cats from dogs.

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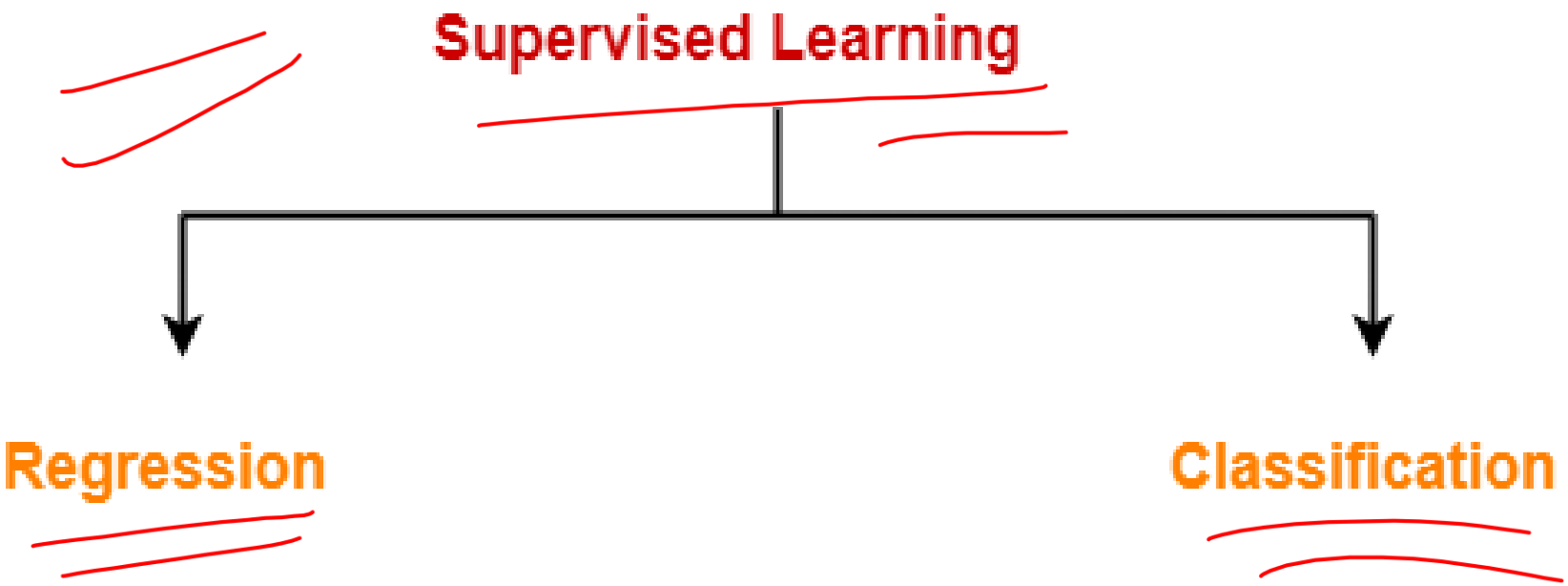
Types of Supervised Learning Algorithm



Example of Supervised Learning

- **Example of Classification:**
  - The spam filter is a good example of this: it is trained with many example emails along with their *class* (spam or ham), and it must learn how to classify new emails.
- **Example of Regression:**
  - Another typical task is to predict a *target* numeric value, such as the price of a house, given a set of *features* (area in square feet, no. of bedrooms etc.) called *predictors*.
  - To train the system, you need to give it many examples of houses, including both their predictors and their labels (i.e., their prices).

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Example of Classification

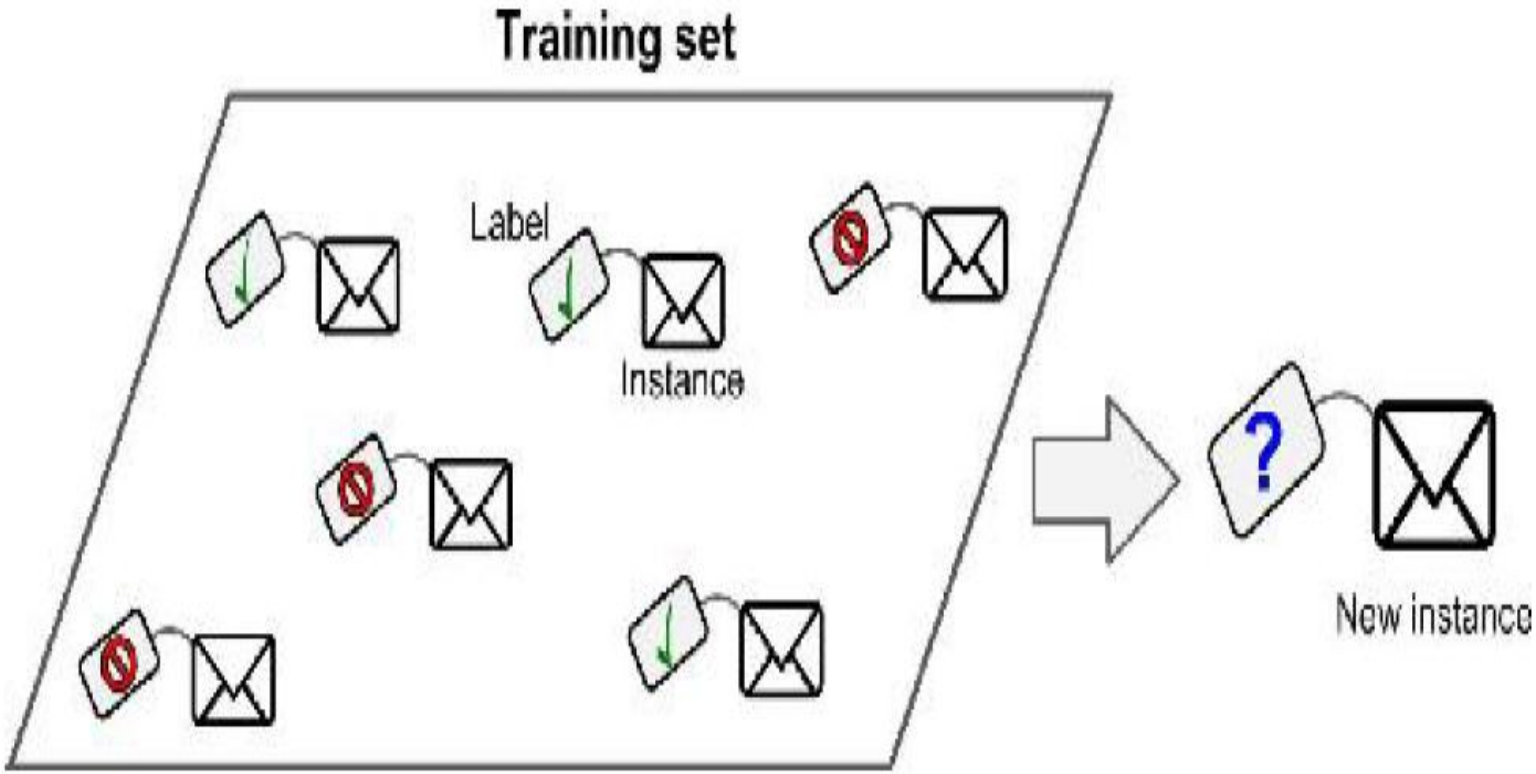

Figure 1-5. A labeled training set for supervised learning (e.g., spam classification)

Application of Classification: Spam Detection

- Spam detection in email service providers can be identified as a classification problem.
- This is binary classification since there are only 2 classes as spam and not spam.
- A classifier utilizes training data to understand how given input variables relate to the class.
- In this case, known spam and non-spam emails have to be used as the training data. When the classifier is trained accurately, it can be used to detect an unknown email.

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Example of Classification: Heart Problem Prediction

Training set

Age	Heart rate	Blood pressure	Heart problem
65	78	150/70	Yes
37	83	112/76	No
71	67	108/65	No

Prediction set

Age	Heart rate	Blood pressure	Heart problem
43	98	147/89	?
65	58	106/63	?
84	77	150/65	?

TABLE 1 – TRAINING AND PREDICTION SETS FOR MEDICAL DATABASE

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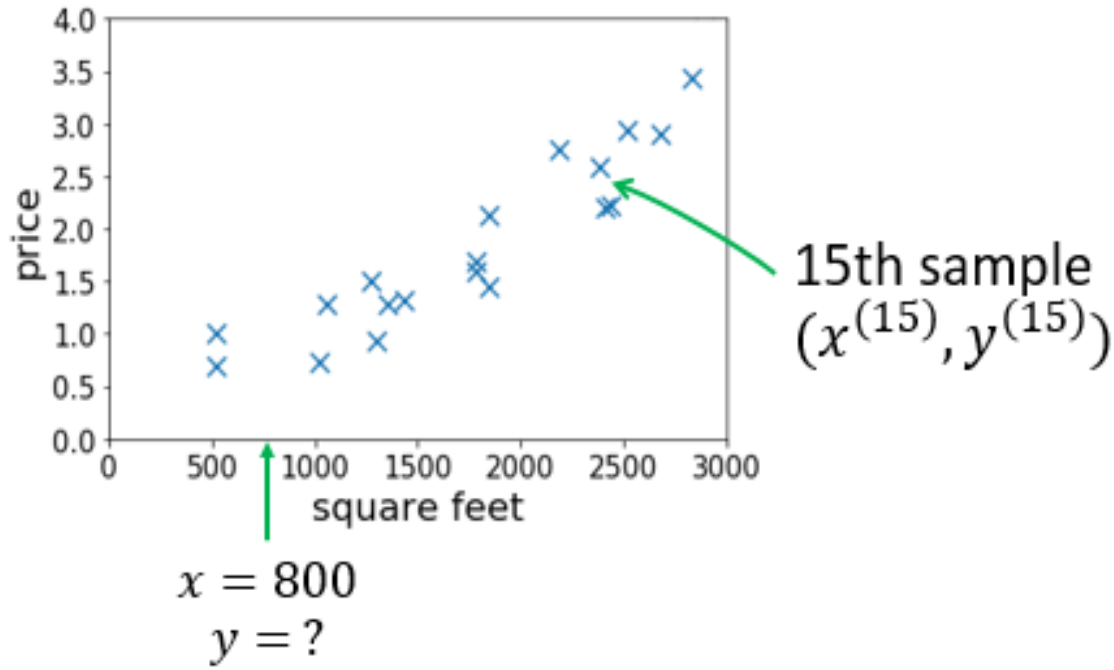
Example of Regression: Housing Price Prediction

- Given: a dataset that contains n samples

$$\left(x^{(1)}, y^{(1)}\right), \ldots \left(x^{(n)}, y^{(n)}\right)$$

- Task: if a residence has x square feet, predict its price?



Supervised Learning Algorithms

- Some of the supervised learning algorithms:
  - k-Nearest Neighbors
  - Linear Regression
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Trees and
  - Random Forests

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Advantages and Challenges of Supervised Learning

- **Advantages:**
  - Highly accurate with sufficient labeled data.
  - Well-understood mathematical models.
- **Challenges:**
  - Requires a large amount of labeled data.
  - Prone to overfitting if the model is too complex.

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Where to Use Supervised Learning

- When you have labeled data and want to predict outcomes.
- Ideal for classification (like spam detection) or regression tasks (like price forecasting).
- Best used in domains where historical data with outcomes is already available.

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Applications of Supervised learning

- Supervised learning is used in a wide variety of applications, including:
  - **Image, speech and text processing:** For tasks like image classification, speech recognition and sentiment analysis.
  - **Predictive analytics:** To forecast sales, customer churn, stock prices and weather conditions.
  - **Recommendation and personalization:** Powering systems that suggest products, movies or content.
  - **Healthcare and finance:** Used for medical diagnosis, fraud detection and credit scoring.
  - **Automation and control:** In autonomous vehicles, manufacturing quality checks and gaming AI.

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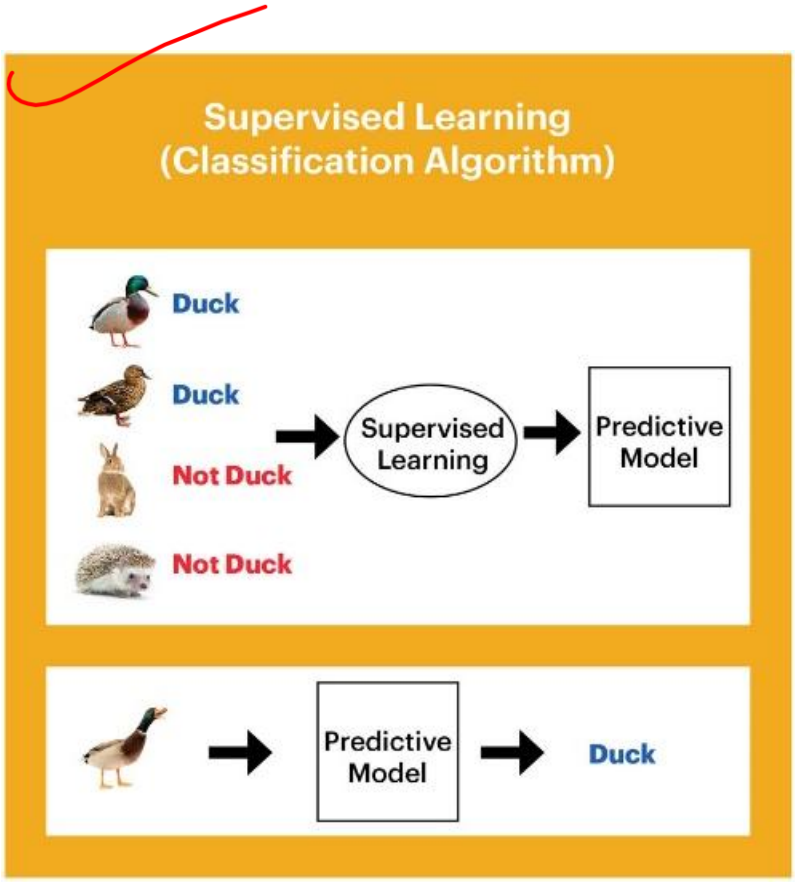
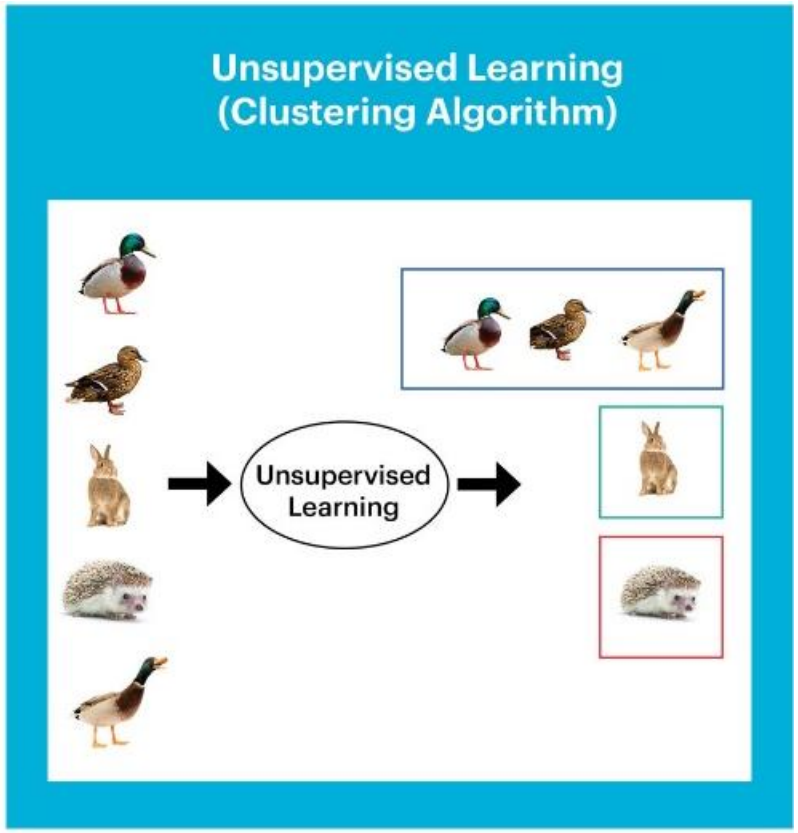
Supervised Learning vs Unsupervised Learning





Unsupervised Learning

- In *unsupervised learning,* the training data is unlabeled (Figure 1-7).
- Here, the algorithm is given a dataset **with only input features (like customer purchase history) but no labels.**
- Then machine tries to find structure, patterns, and relationships in the data without explicit supervision.
- It might group similar data points together or identify trends.
- In this way, it provides insights, **such as clusters of similar data or patterns that were not obvious before.**

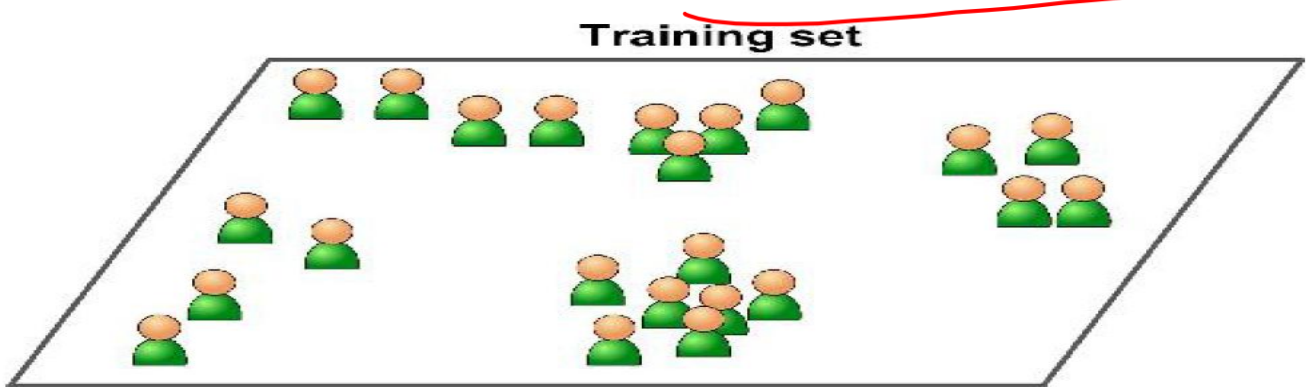
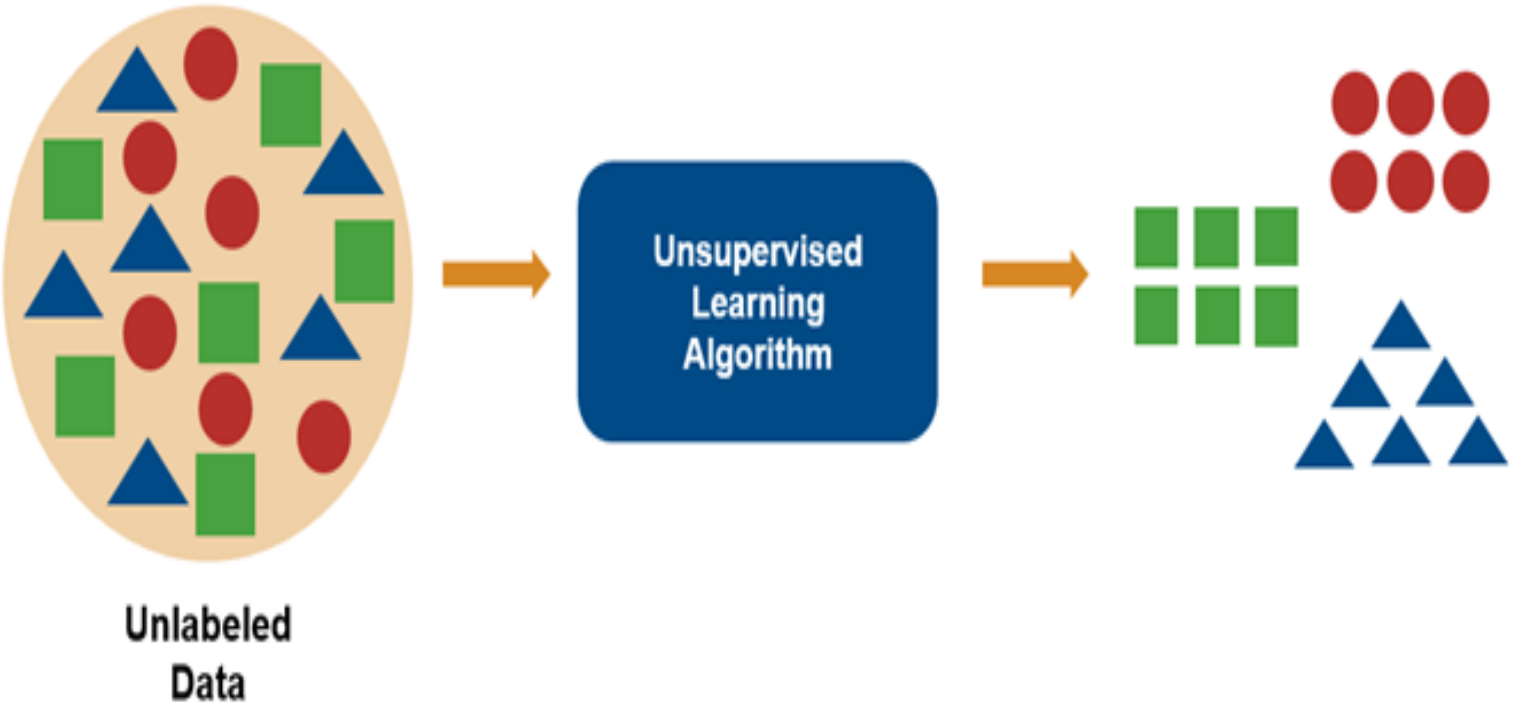


Figure 1-7. An unlabeled training set for unsupervised learning

Unsupervised Learning



Unsupervised Learning Analogy

- **1. Sorting Books Without Labels** : Imagine you are given a box of books with no labels or categories. Your task is to organize them:
  - You notice that some books are mystery novels, so you group them together.
  - Others are textbooks, which you set aside in a separate pile.
  - Comic books form another group because of their distinct style.
  - Here, you create groups based on the books' characteristics (e.g., genre, content) without any prior guidance. This reflects how unsupervised learning clusters data based on similarities.

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Unsupervised Learning

- In this type of machine learning algorithm,
  - The training data set is an unlabelled data set.
  - In other words, the training data set contains only the input value (X) and not the target value (Y).
  - These algorithms try to use techniques on the input data to *mine for rules*, *detect patterns*, and *summarize and group the data points* which help in deriving meaningful insights and describe the data better to the users.
  - Example problems are clustering, dimensionality reduction and association rule learning.

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Types of Unsupervised Learning Algorithm

- There are mainly 3 types of Unsupervised Algorithms that are used:
  - **Clustering**
  - **Association Rule Learning**
  - **Dimensionality Reduction**

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Unsupervised Learning Algorithm: Clustering

- **Clustering**: is an unsupervised machine learning technique that groups unlabeled data into clusters based on similarity.
- Its goal is to discover patterns or relationships within the data without any prior knowledge of categories or labels.
- Used where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behaviour.
- Example: Customer segmentation.

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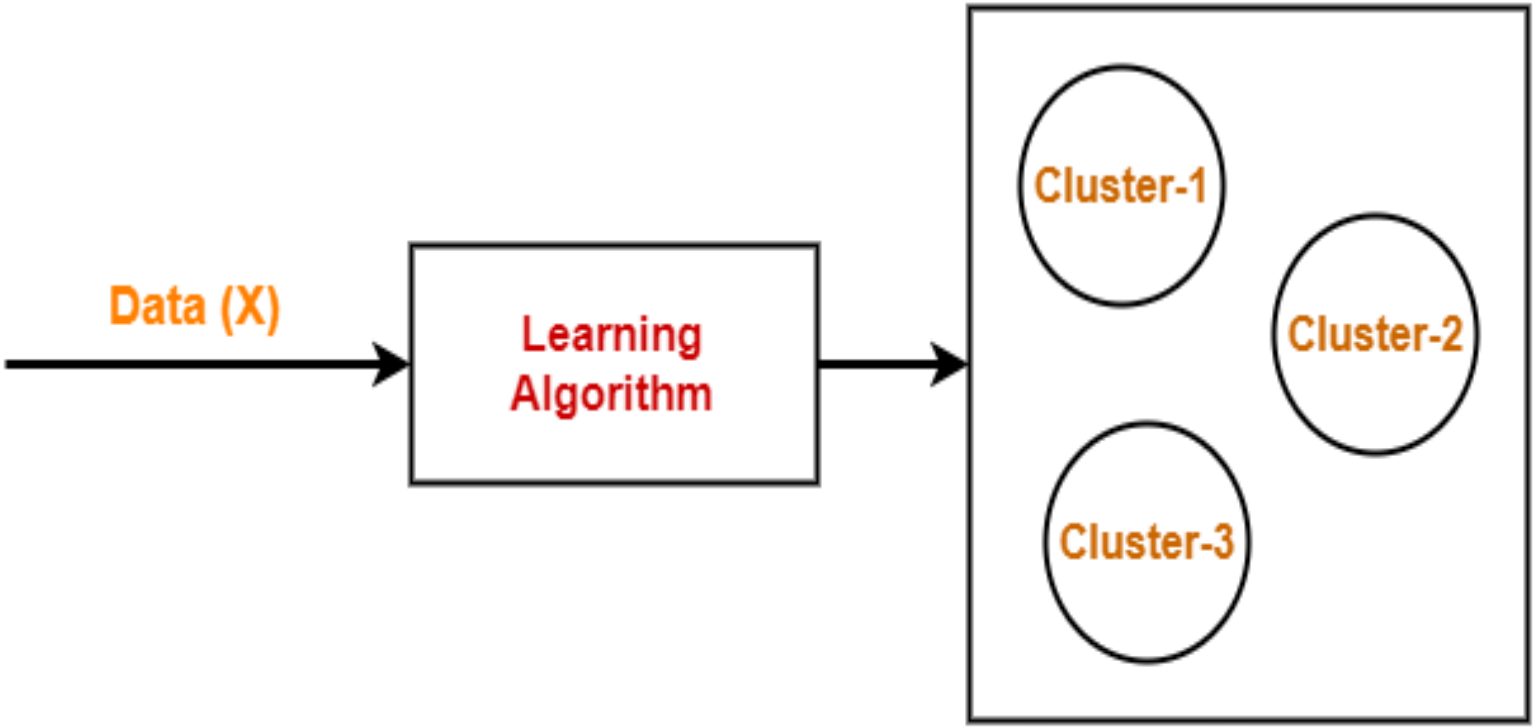
Unsupervised Learning: Clustering



Unsupervised Learning: Clustering

- **For instance**, suppose it is given an image having both dogs and cats which have not seen ever.
- Thus the machine has no idea about the features of dogs and cat so we can't categorize it in dogs and cats.
- But it can categorize them according to their similarities, patterns, and differences i.e., we can easily categorize the above picture into two parts.
- First may contain all pics having **dogs** in it and second part may contain all pics having **cats** in it.
- Here you didn't learn anything before, means no training data or examples.

Unsupervised Learning: Dimensionality Reduction

- **Dimensionality reduction:** the goal is to simplify the data without losing too much information.
- It reduces the number of variables in a dataset while retaining essential information.
- One way to do this is to merge several correlated features into one.
- For example, a car's mileage may be very correlated with its age, so the dimensionality reduction algorithm will merge them into one feature that represents the car's wear and tear. This is called *feature extraction*
  - Principal Component Analysis (PCA) Algorithm

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Unsupervised Learning: Association Rule Learning

- **Association Rule Learning** — is a rule-based unsupervised learning technique used to discover interesting relationships between variables in large datasets.
- It identifies patterns in the form of "if-then" rules, showing how the presence of some items in the data implies the presence of others.
- Example: *Market Basket Analysis*. This means using a database of transactions in a supermarket to find items that are bought together.
- For example, a person who buys potatoes and burgers usually buys beer. This insight could be used to optimize the supermarket layout.
- Online stores usually suggest to you a new item based on the items you have bought. They analyze online transactions to find patterns in the buyer's behavior

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Applications of Unsupervised learning

- **Customer Segmentation**: Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.
- **Anomaly Detection**: Identifies unusual patterns in data, aiding fraud detection, cybersecurity and equipment failure prevention.
- **Recommendation Systems**: Suggests products, movies or music by analyzing user behavior and preferences.
- **Image and Text Clustering**: Groups similar images or documents for tasks like organization, classification or content recommendation.
- **Social Network Analysis**: Detects communities or trends in user interactions on social media platforms.

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Unsupervised Learning: Anomaly detection

- *Anomaly detection*
  - detecting unusual credit card transactions to prevent fraud,
  - catching manufacturing defects, or
  - automatically removing outliers from a dataset before feeding it to another learning algorithm.
- The system is trained with normal instances, and when it sees a new instance it can tell whether it looks like a normal one or whether it is likely an anomaly (see Figure 1-10).

Figure 1-10. Anomaly detection

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Unsupervised Learning Algorithms

- Some of the unsupervised learning algorithms:
  - **Clustering**
    - k-Means
    - Hierarchical Clustering
  - **Association rule learning**
    - Apriori
    - ECLAT
  - **Dimensionality Reduction**
    - Principal Component Analysis (PCA)

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Advantages and Challenges of Unsupervised Learning

- **Advantages:**
  - No need for labeled data, making it cost-effective.
  - Can uncover hidden patterns and insights.
- **Challenges:**
  - Difficult to evaluate model performance.
  - Risk of discovering irrelevant patterns (overfitting).

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Difference between Supervised Learning & Unsupervised Learning

Aspect	Supervised Learning	Unsupervised Learning
Input Data	Uses labeled data (input features + corresponding outputs).	Uses unlabeled data (only input features, no outputs).
Goal	Predicts outcomes or classifies data based on known labels.	Discovers hidden patterns, structures, or groupings in data.
Computational Complexity	Less complex, as the model learns from labeled data with clear guidance.	More complex, as the model must find patterns without any guidance.
Types	**Classification** (for discrete outputs) or **regression** (for continuous outputs).	**Clustering**, **Association, Dimensionality Reduction**
Testing the Model	Model can be tested and evaluated using labeled test data.	Cannot be tested in the traditional sense, as there are no labels.
Algorithms	Linear Regression, Logistic Regression. Support Vector Machines, Decision Trees, Random Forests	K-Means Clustering Hierarchical Clustering. Principal Component Analysis (PCA), Apriori Algorithm

Reinforcement Learning

- **Definition:** A feedback-driven approach where the model learns by interacting with the environment and improving iteratively.
- **How Reinforcement Learning works:**
  - The learning system, called an ***agent*** in this context, can observe the environment, select and perform actions, and
    - get ***rewards*** in return or
    - ***penalties*** in the form of negative rewards
  - It must then learn by itself what is the best strategy, called a ***policy***, to get the most reward over time.
  - A policy defines what action the agent should choose when it is in a given situation.
- **Summary:** In **Reinforcement Learning,** an agent learns to make decisions by receiving "rewards" for correct actions and "penalties" for wrong ones to maximize a long-term goal.

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Key Components of Reinforcement Learning

- **Agent** — The learner or decision-maker.
- **Environment** — The external system with which the agent interacts.
- **Actions** — Choices the agent can make.
- **Rewards** — Feedback from the environment based on actions taken.
- **Policy** — The strategy followed by the agent to take actions.

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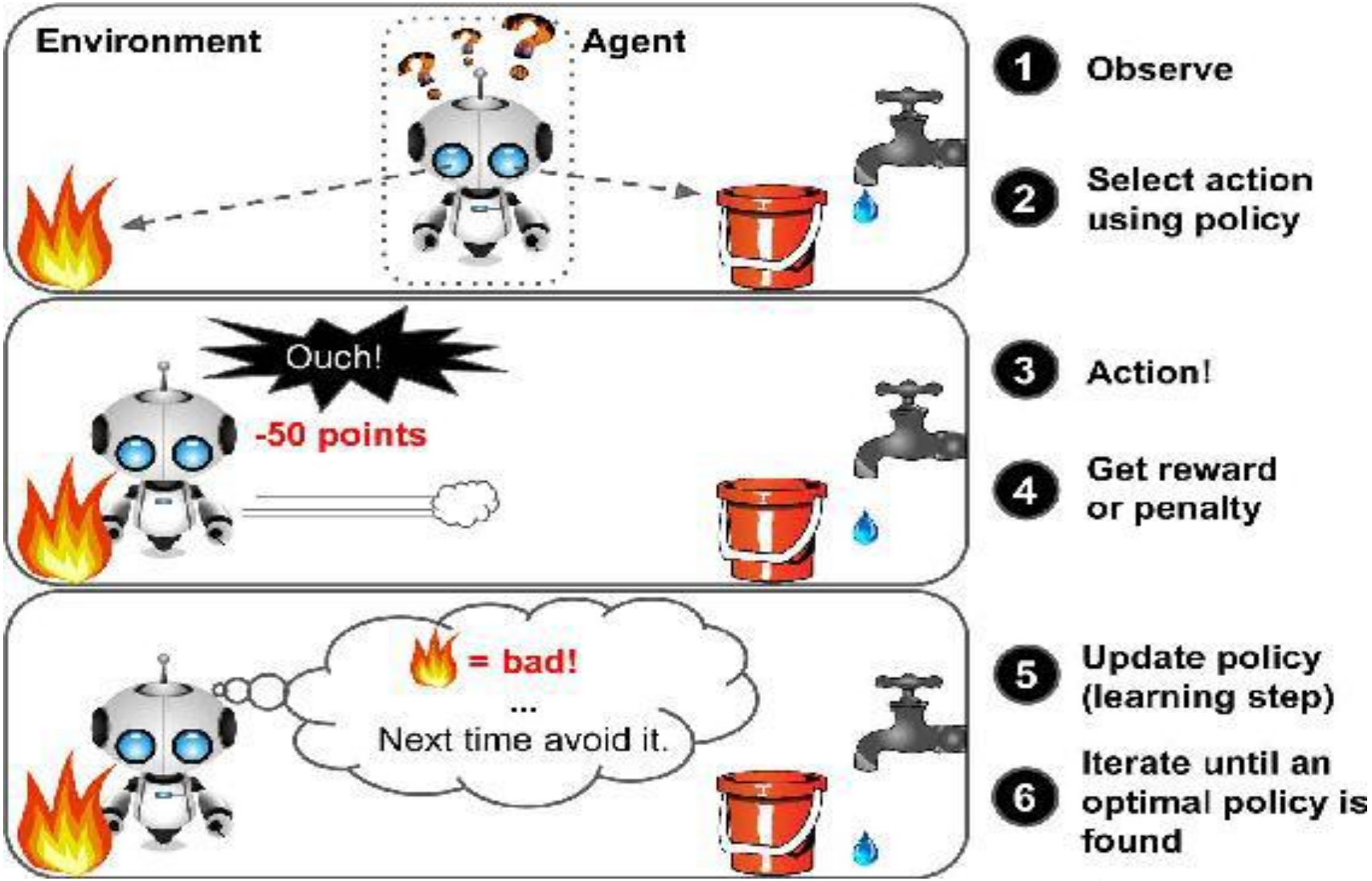
Reinforcement Learning



Examples of Reinforcement Learning

- For example, many **robots** implement Reinforcement Learning algorithms to learn how to walk.
- **DeepMind's AlphaGo program** is also a good example of Reinforcement Learning: it made the headlines in March 2016 when it beat the world champion **Lee Sedol** at the game of ***Go***.
- It learned its winning policy by analyzing millions of games, and then playing many games against itself.
- Note that learning was turned off during the games against the champion; AlphaGo was just applying the policy it had learned.

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Advantages and Challenges od Reinforcement Learning

- **Advantages:**
  - Learns dynamically through interaction.
  - Can optimize complex decision-making processes.
- **Challenges:**
  - Requires significant computational resources.
  - Training can be unstable and require extensive tuning.

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Reinforcement Learning

- Some real-life applications are-
  - Game Playing (board games like Chess)
  - Robot Navigation
  - Self-driving cars

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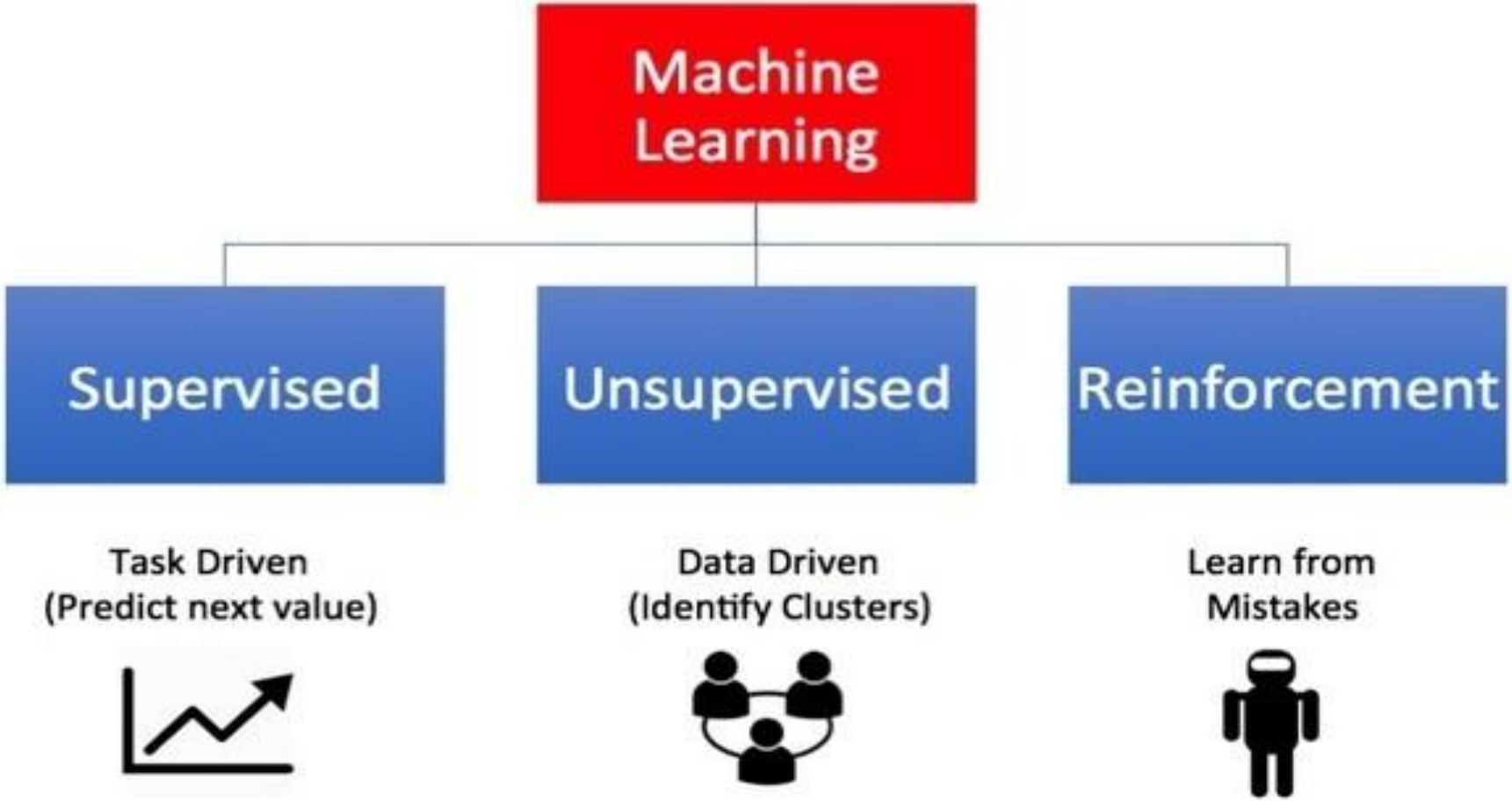
Types of Machine Learning



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Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	Learns from labeled data	Identifies patterns in unlabeled data	Learns through interaction with environment
Type of Data	Labeled data	Unlabeled data	No predefined data learn from environment
Type of Problems	Classification, Regression	Clustering, Association, Dimensionality Reduction	Sequential decision-making
Supervision	Requires external supervision	No supervision	No supervision, learns from feedback
Algorithms	SVM, Decision Trees, Neural Networks	K-Means, PCA, Autoencoders	Q-learning, DQN, SARSA
Goal	Predict outcomes accurately	Discover hidden patterns	Optimize actions for maximum rewards
Applications	Medical diagnosis, Sentiment Analysis	Customer segmentation, Market Basket Analysis	Self-driving cars, robotics, gaming

Real-World Applications

Machine Learning Type	Domain	Examples
Supervised Learning	Healthcare	Disease diagnosis like cancer detection
	Finance	Loan approval, credit risk assessment
	NLP	Sentiment analysis, text classification
Unsupervised Learning	E-commerce	Product recommendation, customer segmentation
	Cybersecurity	Fraud detection, intrusion detection
	Biology	Gene classification, dimensionality reduction
Reinforcement Learning	Autonomous Driving	Self-driving cars learning optimal driving behavior
	Robotics	Training robots for automated assembly tasks
	Gaming	AI-driven strategy games like AlphaGo

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Choosing the Right Learning Approach

- **Supervised Learning:** When labeled data is available for prediction tasks like spam filtering, stock price forecasting.
- **Unsupervised Learning:** When exploring data sets without predefined labels like customer segmentation, anomaly detection.
- **Reinforcement Learning:** When decision-making is required in a dynamic environment like game AI, robotics, self-driving cars.

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